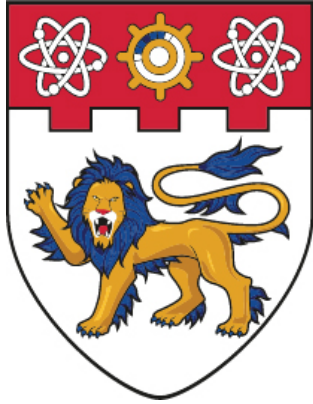




**NANYANG
TECHNOLOGICAL
UNIVERSITY**

SINGAPORE

**Science & Technology for Humanity
CC0007
Assignment 1**

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**Interdisciplinary Collaborative Core Office
2026**

Chosen Theme: AI & Society: Is AI making us less capable of thinking critically?

Part A: Source Library Table

Field	Details
Author/ Year / Title	Lin, J., & Al-Hada, N. M. (2026). The critical-thinking paradox in generative AI-integrated learning: Distinguishing efficiency from cognitive depth—a differentiated framework and testable propositions. <i>Frontiers in Psychology</i> , 17, 1906070. https://doi.org/10.3389/fpsyg.2026.1906070 https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2026.1906070/full
Disciplinary Lens	Social Science
Roles in Knowledge Base (max 30 words)	The improvement of short term tasks by critical offloading onto AI, reducing deep cognitive thinking for long-term critical thinking which provides us insights that AI can be helpful as well.

Field	Details
Author/ Year / Title	Tian, J., & Zhang, R. (2025). Learners' AI dependence and critical thinking: The psychological mechanism of fatigue and the social buffering role of AI literacy. <i>Acta Psychologica</i> , 260, 105725. https://doi.org/10.1016/j.actpsy.2025.105725 https://www.sciencedirect.com/science/article/pii/S0001691825010388
Disciplinary Lens	Social Science
Roles in Knowledge Base (max 30 words)	This source showcases higher AI dependence resulting in lower levels of critical thinking with cognitive fatigue added. This gives us insights on how cognitive fatigue is present in using AI.

Field	Details
Author/ Year / Title	Frenkenberg, A., & Hochman, G. (2025). It's Scary to Use It, It's Scary to Refuse It: The Psychological Dimensions of AI Adoption—Anxiety, Motives, and Dependency. <i>Systems</i> , 13(2), 82. https://doi.org/10.3390/systems13020082 https://www.mdpi.com/2079-8954/13/2/82
Disciplinary Lens	Social Science
Roles in Knowledge Base (max 30 words)	This source showcases participants frequently engaged with AI demonstrating significant motivational factors and anxiety. This gives us insights on how anxiety correlates with the usage of AI.

Field	Details
Author/ Year / Title	European Commission: Directorate-General for Education, Youth, Sport and Culture. (2022). <i>Ethical guidelines on the use of artificial intelligence (AI) and data in teaching and learning for educators</i> . Publications Office of the European Union. https://data.europa.eu/doi/10.2766/153756 .
Disciplinary Lens	Governance
Roles in Knowledge Base (max 30 words)	This source (European Commission) identifies over-reliance on AI systems as a core institutional hazard in classroom and institution-based environments and provides a regional policy framework. It can also lead to algorithmic bias and data privacy violations.

Field	Details
Author/ Year / Title	U.S. Department of Education, Office of Educational Technology. (2023). <i>Artificial intelligence and the future of teaching and learning: Insights and recommendations</i> . https://www2.ed.gov/documents/ai-report/ai-report.pdf
Disciplinary Lens	Governance
Roles in Knowledge Base (max 30 words)	This source (Department of Education) focuses on cognitive offloading, where analytical, instructional, or evaluative tasks are fully transferred to algorithmic models. It directly addresses the risk of automated decision-making displacing human reasoning.

Field	Details
Author/ Year / Title	Department for Education. (2025). <i>Generative AI: Product safety standards</i> . GOV.UK. https://www.gov.uk/government/publications/generative-ai-product-safety-standards
Disciplinary Lens	Governance
Roles in Knowledge Base (max 30 words)	This source provides a government policy perspective on potential cognitive effects of genAI in education. It is relevant as the UK Department for Education identifies cognitive deskilling and offloading as potential risks and established standards to replace learners' cognitive effort.

Field	Details
Author/ Year / Title	Microsoft Research. (2025). <i>Rethinking AI in knowledge work: From assistant to tool for thought</i> . https://www.microsoft.com/en-us/research/articles/rethinking-ai-in-knowledge-work-from-assistant-to-tool-for-thought/
Disciplinary Lens	Business
Roles in Knowledge Base (max 30 words)	AI may result in workers becoming validators of AI's ideas instead of better workers themselves. This source presents a counter where if designed properly, AI can encourage deeper thinking instead.

Field	Details
Author/ Year / Title	Organisation for Economic Co-operation and Development. (2025). <i>Generative AI and the SME workforce: New survey evidence</i> . OECD Publishing. https://www.oecd.org/en/publications/generative-ai-and-the-sme-workforce_2d08b99d-en/full-report/component-5.html
Disciplinary Lens	Business
Roles in Knowledge Base (max 30 words)	This OECD report analyzes how generative AI is being adopted by SMEs based on a 2024 survey with more than 5,000 SMEs across seven countries.

Field	Details
Author/ Year / Title	Gerlich, M. (2025). AI tools in society: Impacts on cognitive offloading and the future of critical thinking. <i>Societies</i> , 15(1), Article 6 https://www.mdpi.com/2075-4698/15/1/6
Disciplinary Lens	Business
Roles in Knowledge Base (max 30 words)	This studies the idea of cognitive offloading as a side effect of AI usage in performing cognitive tasks. It shows the relationship between AI use, cognitive offloading and critical thinking.

Field	Details
Author/ Year / Title	Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. <i>Proceedings of the ACM on Human-Computer Interaction</i> , 5(CSCW1), Article 188, 1–21. https://doi.org/10.1145/3449287

Disciplinary Lens	STEM
Roles in Knowledge Base (max 30 words)	This source demonstrates that cognitive forcing functions, like requiring users to make an initial decision before seeing AI suggestions, significantly reduces acceptance of incorrect AI recommendations compared to explanation-only interfaces.

Field	Details
Author/ Year / Title	Lee, H. P. H., Sarkar, A., Tankelevitch, L., Drosos, I., Rintel, S., Banks, R., & Wilson, N. (2025). The impact of generative AI on critical thinking: Self-reported reductions in cognitive effort and confidence effects from a survey of knowledge workers. In <i>Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems</i> (CHI '25). Association for Computing Machinery. https://doi.org/10.1145/3706598.3713778
Disciplinary Lens	STEM
Roles in Knowledge Base (max 30 words)	This source shows that knowledge workers with higher confidence in AI output report exerting less critical thinking effort across tasks like analysis and evaluation, linking confidence to cognitive disengagement.

Field	Details
Author/ Year / Title	Reichert, L., Zhang, Z. T., von Oswald, E., Liu, Y., Rogers, Y., & Hassib, M. (2025). AI, help me think—but for myself: Assisting people in complex decision-making by providing different kinds of cognitive support. In <i>Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems</i> (CHI '25). Association for Computing Machinery. https://doi.org/10.1145/3706598.3713295
Disciplinary Lens	STEM
Roles in Knowledge Base (max 30 words)	This source reveals that an AI system extending a user's own reasoning, rather than issuing direct recommendations, leads users to integrate AI input more closely and retain greater decision ownership.

Field	Details
Author/ Year / Title	Clark, A., & Chalmers, D. (1998). The Extended Mind. <i>Analysis</i> , 58(1), 7–19. https://www.alice.id.tue.nl/references/clark-chalmers-1998.pdf
Disciplinary Lens	Humanities

Roles in Knowledge Base (max 30 words)	This source provides the philosophical basis regarding cognition, mind, the human self, and the differentiation of cognitive extension as opposed to AI reliance, undermining critical thinking.
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Field	Details
Author/ Year / Title	Vallor, Shannon, The AI Mirror: How to Reclaim Our Humanity in an Age of Machine Thinking (New York, 2024; online edn, Oxford Academic, 30 Apr. 2024), https://doi.org/10.1093/oso/9780197759066.001.0001 , accessed 1 Sept. 2026.
Disciplinary Lens	Humanities
Roles in Knowledge Base (max 30 words)	This source examines the role of AI, the past, contemporary, and near future interactions with AI systems, offering a balanced perspective on the various challenges of coexisting with AI.

Field	Details
Author/ Year / Title	Su, J. (2024). Consciousness in artificial intelligence: A philosophical perspective through the lens of motivation and volition. <i>Critical Debates in Humanities, Science and Global Justice</i> , 3(1). https://criticaldebateshsgj.scholasticahq.com/article/117373-consciousness-in-artificial-intelligence-a-philosophical-perspective-through-the-lens-of-motivation-and-volition
Disciplinary Lens	Humanities
Roles in Knowledge Base (max 30 words)	This source examines consciousness and breaks it down to volition and motivation, and why AI fulfils volition, but cannot achieve true motivation without first becoming conscious.

Part B: Problem Statement (200-250 words)

Based on your research, write a concise summary addressing:

What is the prevalence and significance of the problem you have chosen to investigate?

- What does the current literature tell you about this problem?
- What is the gap you want to address (i.e. what remains unresolved, underexplored, or poorly understood?)

Many businesses, workplaces and educational settings have adopted Generative AI where several places are using AI as tools. There is evidence of critical thinking degradation when dependent on AI. There is also a long-term behavioral impact where there is a retention drop in knowledge, not doing well in tests and weakening functional brain connectivity due to assistance and dependency of AI. It can create cognitive fatigue and false illusions that by accepting by default what AI recommends. (77 Words)

Current literature converges on several reasons for cognitive function degradation. Generative AI provides an easy way to generate polished products to users, allowing humans to offload mental tasks onto AI to conserve cognitive resources. Hence, they bypass the cognitive processes required to develop critical thinking skills. High trust towards AI and the mental fatigue of AI work validation leads to passively accepting automated suggestions and suppressing independent reasoning. The illusion of competence further lowers awareness of the need to think critically. (81 words)

The gaps to address are conflicting literature findings, methodological limitations causing constrained conclusions and unaddressed tech solutions and policies. Current literature conflicts on whether AI aids learning. While AI boosts immediate output quality, users prefer direct answers over effortful tools, causing unaided reasoning to decline. The sources explicitly outline methodological limitations such as reliance on self-reported data, unverified neurophysiological models, and demographic biases toward younger technologically proficient students. Current policies and technologies prioritise immediate output, commercial frictionless design, and unsustainable human validation, ignoring the long-term degradation of independent critical thinking skills. (91 words)

Total: 249 words

Part C: Critical AI Reflection

Insight:

Failure:

Intervention: